COVID-19 Early Case Trend Analysis & Recovery Insights

1. Introduction

This project is based on a case study provided by HealthGuard Analytics Pvt. Ltd., a healthcare data analytics firm that delivers insights to government health departments and hospitals.

The objective of this study is to analyze early-stage COVID-19 patient data to understand infection patterns, demographic characteristics, recovery behavior, and regional trends. These insights are intended to support data-driven public health decisions.

2. Business Context

HealthGuard Analytics collaborated with a national public health authority to analyze patient-level data from the early phase of a COVID-19 outbreak.

The dataset includes confirmed cases along with demographic details, infection history, exposure metrics, and patient outcomes such as released, isolated, or deceased.

The analysis aims to improve screening strategies, isolation planning, and early outbreak response systems.

3. Problem Statement

The public health authority seeks answers to the following key questions:

1. Who is getting infected?
2. How are infections spreading?
3. What are the recovery trends?
4. Which regions are most impacted?
5. What factors influence recovery time?

4. Analytical Objectives

- Perform Exploratory Data Analysis (EDA)
- Apply descriptive statistics
- Visualize trends (gender, age, region, infection sources, recovery timelines)
- Calculate recovery duration
- Apply Linear Regression to predict recovery time

5. Dataset Description

Demographics

- Sex, Birth Year, Country, Region

Infection Details

- Infection Reason, Infection Order, Infected By

Exposure Metrics

- Contact Number

Timeline Data

- Confirmed Date, Released Date, Deceased Date

Case Outcome

- State (Released, Isolated, Deceased)

6. Methodology

Data Cleaning

- Converted date columns into datetime format
- Created age from birth year
- Handled missing values

- Created separate datasets for analysis

Feature Engineering

- Created recovery_days
- Generated age groups

Exploratory Data Analysis

- Gender and age distribution
- Regional analysis
- Infection source analysis

Recovery Analysis

- Distribution of recovery time
- Age and gender comparison
- Correlation analysis

Machine Learning

- Linear Regression
- Random Forest Regressor
- Model comparison

7. Results and Findings

Demographics

- Most affected age group: 36–60 years
- Gender distribution is balanced

Regional Impact

- Highest cases in Gyeongsangbuk-do and Capital Area

Infection Spread

- Major sources:
    - Contact with infected individuals
    - Travel-related exposure

Recovery Trends

- Average recovery time ≈ 15 days
- Most recover within 7–24 days

8. Model Results and Evaluation

8.1 Dataset Summary

- EDA dataset size: 272 records
- Recovery dataset size: 28 records
- ML dataset size: 28 records

8.2 Linear Regression Results

- R^2 Score: -0.2065
- Mean Absolute Error: 5.15

Interpretation:
The model performs poorly, as indicated by the negative R^2 score. Predictions deviate by approximately 5 days.

Coefficients:

- Intercept: 17.88
- Age: -0.0766

- Sex: 0.8566

Insight:

Age and gender have weak influence on recovery time.

8.3 Random Forest Results

- R^2 Score: -1.6634
- Mean Absolute Error: 8.28

Interpretation:

Random Forest performs worse than Linear Regression in this case.

8.4 Feature Importance

Feature	Importance
Age	0.877
Sex	0.123

Insight:

Age is the dominant factor influencing recovery.

8.5 Model Conclusion

- Linear Regression outperformed Random Forest
- Model performance is limited due to small dataset
- Results are exploratory

9. Expected Outcomes

- Clear understanding of infection patterns
- Identification of high-risk groups
- Insights into recovery timelines
- Visual support for decision-making

10. Optional Extensions

- Feature engineering for recovery duration
- Linear Regression modeling
- Model evaluation using R^2 score

11. Limitations

- High missing values
- Small recovery dataset
- Limited predictive power

12. Conclusion

This project demonstrates how data analytics and machine learning can be applied to understand infectious disease trends.

Despite data limitations, meaningful insights were obtained regarding infection spread, demographic impact, and recovery behavior. These findings can support better public health strategies.

13. Author

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